Multi-Cluster Baseline: From 41% to 100% Drift Recall

Summary

The single biggest improvement in the drift detector came from changing how "normal" traffic is represented — not from a new detection algorithm, but from replacing a single average point with several. This one change took measured drift-cohort recall from 41% to 100% on the labeled synthetic dataset, with false-positive rate moving only slightly (roughly 5% to 6%). This page documents why the original approach failed, what replaced it, and how the improvement was measured.

The Original Design: One Centroid Per Route

The drift detector's baseline job computes what "normal" traffic looks like for each route, then scores new traffic by distance from that baseline (see the Mahalanobis Distance page for the distance calculation itself). The original implementation defined "normal" as a single centroid — the mean embedding vector across all normal-cohort traffic for that route — plus a per-dimension variance for the distance calculation.

Why a Single Centroid Failed

Real "normal" traffic for a given route is not one thing. Our synthetic `normal` cohort alone spans multiple distinct topics — general Q&A, coding questions, summarization requests — and real production traffic would be at least as varied. Averaging all of that into one centroid produces a point that doesn't actually sit near any of the real topic clusters; it lands in the empty space between them.

The practical consequence: a completely legitimate prompt on a less common topic could score as "far from normal" — not because anything about it was actually anomalous, but because the single averaged centroid simply didn't represent that topic well. This capped detection performance in a specific way: the threshold had to be loose enough to avoid flagging legitimate topic variation, which meant it was also too loose to reliably catch real drift. Measured against the labeled synthetic drift cohort, this setup caught only **41%** of drift sessions.

The Fix: K-Means Sub-Clusters

Instead of one centroid per route, the baseline job now:

1. Runs k-means clustering over the route's normal-cohort embeddings, producing **four sub-clusters** instead of one average.
2. Computes a centroid and per-dimension variance for each sub-cluster independently.
3. Scores a new prompt by its distance to the **nearest** sub-cluster — not by distance to a single blended average.

This is the same k-means mechanism already used to power the dashboard's Topic Clusters page, so the multi-cluster baseline reuses an existing computation rather than introducing new infrastructure.

Conceptually, this replaces the question "how far is this from the average of everything normal?" with "how far is this from the closest thing to it that we know is normal?" — a strictly more accurate model of what normal traffic actually looks like when it's genuinely multi-topic.

Measured Impact

Both numbers below were measured the same way: run the detector against the full labeled synthetic dataset, and compare its output directly against the known ground-truth `cohort` column — not eyeballed, not estimated.

	Single centroid	Four sub-clusters
Drift-cohort recall	41% (33/80)	**100% (80/80)**

False-positive rate (normal cohort)	~5%	~6%

The false-positive increase is small and expected — a nearest-cluster comparison is inherently a slightly looser match than a single fixed point, so a marginal rise in false positives is the honest cost of this fix. The recall gain overwhelmingly outweighs it.

Why This Was Catchable at All

This entire investigation was only possible because the synthetic dataset carries ground-truth `cohort` labels. Without a known-correct answer to check against, a detector catching "some" drift sessions could easily look adequate on a surface pass — the failure only becomes visible when you can compute an exact recall number against labeled data. See the Synthetic Dataset & Ground-Truth Validation page for more on this.